

Syllabus

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Course Title

Statistics for Genomic Data Science

Course Instructor(s)

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Course Description

It isn't enough to just know how to use the tools. Doing genomic data science well means understanding the statistical principles at work. This class will provide an introduction to the statistics behind the most popular genomic data science projects. This will help you ask better questions, plan better research, and interpret the results more accurately.

Course Content

Introduction

- Welcome
- What is statistics?
- Finding statistics you can trust
- Getting help
- What is data?
- Representing data

Week 1

- Week 1 introduction
- Reproducible Research
- Achieving reproducible research
- R markdown
- The three tables in genomics
- The three tables in genomics (in R)
- Experimental Design: variability, replication, and power
- Experimental Design: confounding and randomization
- Exploratory Analysis
- Exploratory Analysis in R

- Data transforms
- Clustering
- Clustering in R

Week 2

- Week 2 Introduction
- Dimension reduction
- Dimension reduction (in R)
- Pre-processing and normalization
- Quantile normalization (in R)
- The linear model
- Linear models with categorical covariates
- Adjusting for covariates
- Linear regression in R
- Many regressions at once
- Many regression in R
- Batch effects and confounders
- Batch effects in R

Week 3

- Week 3 introduction
- Logistic regression
- Regression for counts
- GLMs in R
- Inference
- Null and alternative hypotheses
- Calculating statistics
- Comparing models
- Calculating statistics in R
- Permutation
- Permutation in R
- P-values
- Multiple testing
- P-values and multiple testing in R

Week 4

- Week 4 introduction
- Gene set analysis
- More enrichment
- Gene set analysis in R
- The process for RNA-seq
- The process for Chip-Seq
- The process for DNA methylation
- The process for GWAS/WGS
- Combining data types (eQTL)
- eQTL in R
- Researcher degrees of freedom

- Inference vs. prediction
- Knowing when to get help
- Course Wrap-Up

Lecture Materials

Lecture videos will be released at the beginning of the course and will be available thereafter. You are welcome to view them at your convenience. Accompanying each video lecture will be a PDF copy of the slides. Some lectures also include some R code.

Companion website and R package

You can find all the course lecture materials in the course pages or from the companion site:

- Course materials: http://jtleek.com/genstats_site/

The course R package that contains all the tutorials for the R coding examples during the course.

- Course R package: <https://github.com/jtleek/genstats>

Soft pre-requisites

This class assumes a basic knowledge of genomics and computing that you can find in the courses:

- [Introduction to Genomic Technologies](#)
- [R programming](#)
- [Bioconductor for Genomic Data Science](#)

Quizzes

There are four quizzes. Quiz 1 is due at the end of the first week, Quiz 2 is due at the end of the second week, Quiz 3 is due at the end of the third week, and Quiz 4 is due at the end of the fourth week. See the Quiz pages for exact due dates and hard deadlines.

Quiz Scoring

You may attempt each quiz up to 3 times. The score from your most successful attempt will count toward your grade.

Hard deadlines and soft deadlines

The due date for each quiz is available on the Quizzes page. For Quizzes 1-3, the hard deadline is five days after the due date. Each day late will incur a 10% penalty, but if you use a late day, the penalty will not be applied to that day. There is no grace period for Quiz 4, and you may not use late days to extend the Quiz 4 due date.

Late Days for Quizzes

You are permitted 5 total late days for Quizzes 1-3. Using a late day moves the due date forward one day so you can avoid late submission penalties. You may not use late days for Quiz 4.

Grading policy

- Quiz 1 = 25%
- Quiz 2 = 25%
- Quiz 3 = 25%
- Quiz 4 = 25%

You must receive a final grade of 70% or better to pass the course. If you are enrolled in Signature Track and pass the course, you will receive a course certificate. Signature Track participants who earn a grade of 90% or better will receive a Distinction designation on their course certificate. If you are not enrolled in Signature Track, your grade will be recorded on your Accomplishments page, but you will not be issued a statement of accomplishment.

Typos

- We are prone to a typo or two - please report them and we will try to update the notes accordingly.
 - In some cases, the videos may still contain typos that have been fixed in the lecture notes. The lecture notes represent the most up-to-date version of the course material.
 - If you use Github you can send pull requests on lecture materials to:
 - <https://github.com/jtleek/genstats>
 - https://github.com/jtleek/genstats_site
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Differences of opinion

Keep in mind that currently data analysis is as much art as it is science - so we may have a difference of opinion - and that is ok! Please refrain from angry, sarcastic, or abusive comments on the message boards. Our goal is to create a supportive community that helps the learning of all students, from the most advanced to those who are just seeing this material for the first time.

Plagiarism

Johns Hopkins University defines plagiarism as "...taking for one's own use the words, ideas, concepts or data of another without proper attribution. Plagiarism includes both direct use or paraphrasing of the words, thoughts, or concepts of another without proper attribution." We take plagiarism very seriously, as does Johns Hopkins University.

We recognize that many students may not have a clear understanding of what plagiarism is or why it is wrong. Please see the following guide for more information on plagiarism:

<http://www.jhsph.edu/academics/degree-programs/master-of-public-health/current-students/JHSPH-ReferencingHandbook.pdf>

It is critically important that you give people/sources credit when you use their words or ideas. If you do not give proper credit -- particularly when quoting directly from a source -- you violate the trust of your fellow students.

The Coursera Honor code includes an explicit statement about plagiarism:

I will register for only one account. My answers to homework, quizzes and exams will be my own work (except for assignments that explicitly permit collaboration). I will not make solutions to homework, quizzes or exams available to anyone else. This includes both solutions written by me, as well as any official solutions provided by the course staff. I will not engage in any other activities that will dishonestly improve my results or dishonestly improve/hurt the results of others.

Technical Information

Regardless of your platform (Windows or Mac) you will need a high-speed Internet connection in order to watch the videos on the Coursera web site. It is possible to download the video files and watch them on your computer rather than stream them from Coursera and this may be preferable for some of you.

Here is some platform-specific information:

Windows

The Coursera web site seems to work best with either the Chrome or the Firefox web browsers. In particular, you may run into trouble if you use Internet Explorer. The Chrome and Firefox browsers can be downloaded from: _Chrome: <http://www.google.com/chrome> _Firefox: <http://www.mozilla.org>

Mac

The Coursera site appears to work well with Safari, Chrome, or Firefox, so any of these browsers should be fine.

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